

INLAY: Gradient-Free Knowledge Editing with an External Addressable Memory, and the Limits of Scope Evaluation

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Abstract

Knowledge editing corrects a fact inside a trained language model without retraining. The dominant approach locates the weights responsible for a fact and rewrites them, which works but permanently mutates state shared with everything else the model knows. We take the opposite position: INLAY leaves the model entirely frozen and stores corrections in an external addressable memory, replaying them at decoding time by adding a bias along the intended token’s unembedding direction. Writing an edit is a table insertion — about 5 ms with no gradient step, roughly $1600\times$ cheaper than gradient-based editors — deletion is exact, and when no edit applies the model’s output is unchanged bitwise, so locality holds by construction rather than by measurement. On CounterFact with GPT-J-6B, INLAY reaches 100% edit accuracy against 89.1% for the strongest weight editor, and it extends to unstructured and LLM-extracted evidence, input conditions for which weight-editing methods have no mechanism.

We then report a negative result about the scope decision that this entire architecture family depends on. Executing every candidate action on 1,689 queries to obtain per-query ground truth, we find that an oracle router choosing the best action every time is *exactly* equal to a one-line static policy in all nine dataset-by-condition cells: the maximum attainable gain of any per-query routing method is 0.00 points. Abstention is the sole winning action zero times out of 1,689. The cause is structural — counterfactual benchmarks ask for the post-edit answer, so answering from parametric knowledge is wrong by construction. Current benchmarks therefore cannot reward the scope classifier that SERAC-descendant designs are built around, and reported routing gains on them measure how much harm a gate was doing rather than any decision quality.

1 Introduction

A language model’s knowledge is not stored anywhere one can point to. There is no row reading “the CEO of X is Y ”; the fact exists as a pattern distributed over billions of weights, each of which also participates in thousands of unrelated facts. Every method in this area must deal with that premise.

The dominant family — ROME [9], MEMIT [10], AlphaEdit [2] — deals with it by locating the weights most responsible for a fact and solving for a minimal update. Causal tracing, which corrupts subject tokens and then restores internal activations one site at a time, shows recovery concentrating in mid-layer MLPs at the final subject token. The localisation is real, but every edit permanently mutates shared weights, and the damage compounds under repeated editing. Hase et al. [4] further show that editing at layers causal tracing does *not* highlight often works equally well, which weakens the inference from localisation to edit site.

We ask what happens if the weights are never touched at all.

Contributions. (i) INLAY, a gradient-free editor: frozen model, external addressable memory, logit-space playback, with exact deletion and locality by construction (§3, §5). (ii) Evaluation across all three AKEW input conditions, including evidence formats for which weight editing is inapplicable rather than merely worse (§5). (iii) A diagnosis and fix for multi-hop chain termination, more than tripling accuracy and holding at two model scales (§5.3). (iv) A negative result: on 1,689 queries with per-action ground truth, an oracle router ties a one-line static policy exactly, and abstention is never uniquely correct. Current benchmarks cannot measure the scope decision this architecture family requires (§6).

2 Related Work

Weight editing. ROME [9] treats a mid-layer MLP as a linear associative memory and solves for a rank-one update; MEMIT [10] spreads many updates across several layers; AlphaEdit [2] projects the update into the null space of preserved knowledge. All three require a clean (prompt, target) pair and mutate shared parameters.

Memory-based and in-context editing. SERAC [12] introduces a scope classifier deciding whether an edit applies, routing in-scope inputs to a counterfactual model. MEND [11] learns a gradient transform. IKE [18] shows few-shot demonstrations can teach a model to prefer an injected fact over its parametric belief. GRACE [3] and WISE [15]

add discrete codebooks and side memories respectively. **We build directly on SERAC’s scope-classifier idea and IKE’s demonstration mechanism; neither is claimed as novel here.** Our addressing follows product-key memory [8] and nearest-neighbour language models [7].

Benchmarks. CounterFact and zsRE evaluate single-fact edits; RippleEdits [1] adds entailed consequences; MQuAKE [19] adds multi-hop chains; AKEW [17] supplies unstructured and extracted evidence alongside clean triples. Our §6 is a claim about what this family of benchmarks can and cannot measure.

3 Method

3.1 Design

The model is frozen. Each edit is a row holding a key (an embedding of the fact’s question form), the answer’s token sequence, and metadata. Three properties follow directly (Figure 1a):

- **Writing is insertion**, not optimisation. No gradients, no covariance statistics; measured at 5–15 ms.
- **Deletion is exact**. Removing the row removes the edit. Unlearning a gradient-installed fact remains an open problem, which matters when removal is a regulatory obligation.
- **Edits cannot interfere**. They are physically separate state.

3.2 Addressing

Keys come from a sentence encoder [13] rather than the base model’s hidden states: it is fast, its geometry is purpose-trained for paraphrase matching, and it stays fixed if the base model is swapped. Keys are compressed with a Johnson–Lindenstrauss random projection [6].

The projection is random rather than learned by design. A learned projection would require data, would drift as the edit distribution changed, and would silently invalidate every stored key. A fixed random matrix never invalidates the memory, at the cost of a few points of distortion — the correct side of that trade for an append-only store.

3.3 Gating and playback

Retrieval always returns *something*. A gate stacks an absolute similarity threshold, a margin over the runner-up, and a relation-residual check asking whether the query concerns the stored relation rather than merely the stored subject.

When the gate fires, playback walks the stored token sequence and, at each decoding step, adds a bias along that token’s unembedding column so it wins the argmax (Figure 1b). Positions outside the answer span are untouched.

Intervening in logit space rather than hidden state is a safety argument. A logit bias moves exactly one score; a hidden-state edit propagates through the unembedding to every token’s logit in ways that are hard to bound. And because the intervention is gated, a silent gate leaves behaviour bitwise unchanged.

4 Experimental Setup

Base models: GPT-J-6B, Qwen2.5-7B, Qwen2.5-1.5B-Instruct, Mistral-7B. Benchmarks: CounterFact, zsRE, RippleEdits [1], and AKEW [17] over CounterFact, WikiUpdate and MQuAKE-CF in structured, unstructured and extracted conditions. Baselines run through EasyEdit [14].

Two commitments are worth stating because they cost us results. Splits are **subject-disjoint**: a subject never crosses a train/test boundary, so a learned component cannot memorise an entity and appear to generalise. Scoring uses **one convention everywhere** — diacritic- and case-insensitive substring match against gold and its aliases — rather than each method’s internal metric, so numbers are comparable across methods that report differently.

One implementation detail materially affects correctness: `BaseEditor.edit(sequential_edit=False)` restores weights before returning, so generating after it returns scores the *unedited* model. All baselines here use `sequential_edit=True` with an explicit state-dict snapshot and restore under harness control.

5 Results

5.1 Edit accuracy and write cost

Figure 2 gives the comparison. AlphaEdit is the strongest weight editor at 89.12%, consistent with its null-space constraint making each update more surgical. ROME and MEMIT tie at 83.67%, which is plausible rather than suspicious: single-fact CounterFact is where both converge, and MEMIT’s advantage is multi-edit preservation.

Two results are failures of methods that work elsewhere. **WISE reaches 66.67% despite its edits landing**: EasyEdit’s own post-edit metric reports success on 141 of 147 edits, yet generation-time accuracy falls well below ROME’s. WISE routes through a side-memory module at generation time, and that routing is an additional failure point in-place updates lack. The edit registers; it is not reliably retrieved. **GRACE does not land at all**, reading exactly 0.0 on every edit and tripping a fail-loud guard rather than reporting a misleading number; an independent generation-based harness agreed that its radius-based matching rarely fires on paraphrases.

Write cost is the systems argument. INLAY writes in 5–15 ms with no gradient step against seconds for gradient-based editors, a ratio of roughly 1600×. Inserting a row is not a cheaper optimisation; it is the absence of one.

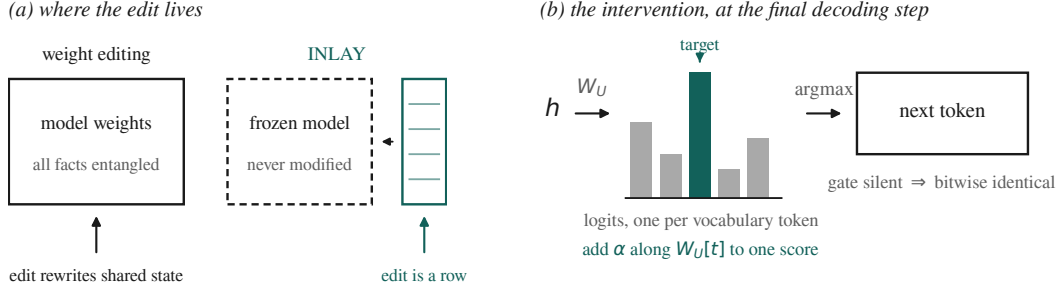


Figure 1: (a) Weight editing rewrites state shared by every fact. INLAY leaves the model frozen and appends a row to a table beside it. (b) The intervention is a bias along one token’s unembedding direction at the final decoding step, so it moves exactly one score. When the gate does not fire, nothing is added and output is bitwise identical.

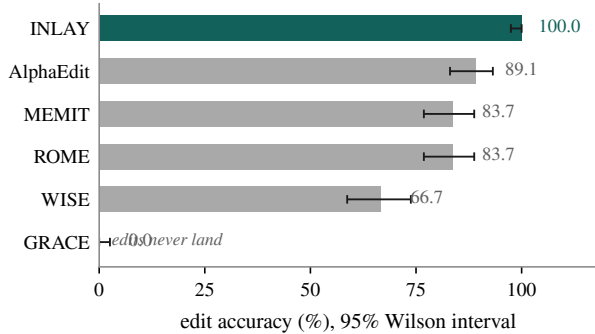


Figure 2: Edit accuracy on CounterFact, structured input, GPT-J-6B, $n = 147$, with 95% Wilson intervals [16]. Scored with a single convention rather than each method’s internal metric.

5.2 Non-structured evidence

Weight-editing methods need a (prompt, target) pair to compute an update from. They have no mechanism for “here is a paragraph of evidence, install whatever matters,” so on AKEW’s unstructured and extracted conditions they are not merely worse but inapplicable. We state this as a structural limitation rather than working around it.

INLAY handles all three conditions. Accuracy is lower on extracted input (78.23%) than on unstructured prose (87.07%) despite retrieval finding the correct card 98.64% of the time in both, which locates the loss in extraction noise rather than retrieval.

5.3 Multi-hop

Our first iterative loop scored 5.0% against a naive single-shot baseline’s 22.5% — four times *worse* than doing nothing. Every sampled failure produced no answer, indicating termination rather than inaccuracy.

The cause was an assumption. MQuAKE-CF chains mix edited facts with ordinary unedited world facts: 277 of 354 groups contain exactly one edit across a two- or three-hop chain. The loop treated every hop as requiring retrieval against the edit index, so a hop needing a never-edited fact terminated the chain answerless. Treating a retrieval miss as a signal to answer that hop from parametric knowledge

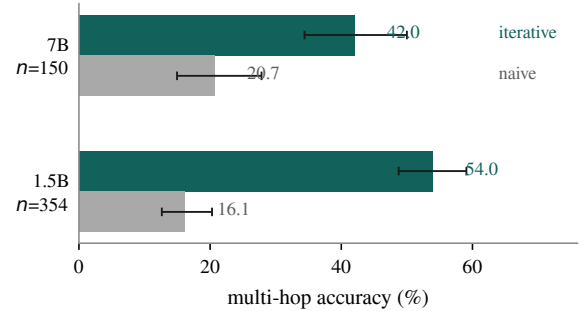


Figure 3: Iterative decomposition versus naive single-shot on MQuAKE-CF after the per-hop fallback fix, with 95% Wilson intervals.

and *continue* raised accuracy to 47.5% on the same sample, and it holds at scale (Figure 3).

6 The Limits of Routing on Current Benchmarks

Every method in this family includes a **scope decision**: given a query, does a stored edit apply, and what should be done about it? We built increasingly careful machinery for it, measured statistically significant gains, and then found the gains were not what they appeared.

6.1 A second-order signal

Gating was net-negative on MQuAKE-CF in every input condition, by up to 15.87 points. Two natural fixes failed diagnostically: recalibrating the verifier threshold moved the false-fire rate from 18.33% to 18.22%, and raising the confidence threshold from 0.85 to 0.97 changed *nothing* — byte-identical decisions, because confidence on the *wrong* retrievals already exceeded 0.97.

The router was asking how confident the verifier was, when it needed to know whether that confidence was *discriminative*. A verifier scoring 0.99 on the top candidate and 0.98 on four unrelated ones is saturated, not confident, and no threshold on a top-1 score can see that.

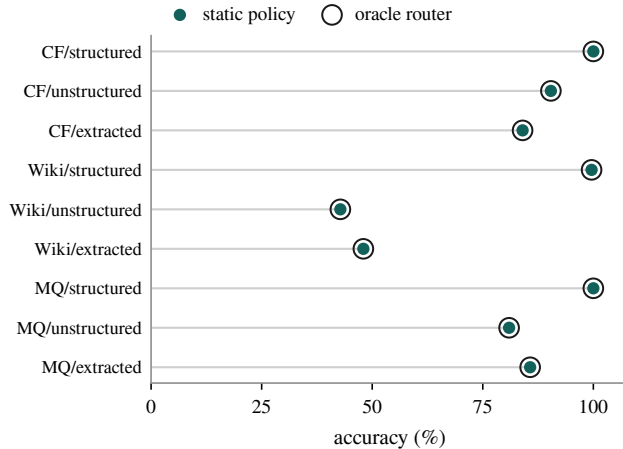


Figure 4: Every candidate action executed and scored on 1,689 queries (CF = CounterFact, MQ = MQuAKE-CF). The oracle ring is concentric with the static-policy point in all nine cells: the best attainable router and one line of code are indistinguishable.

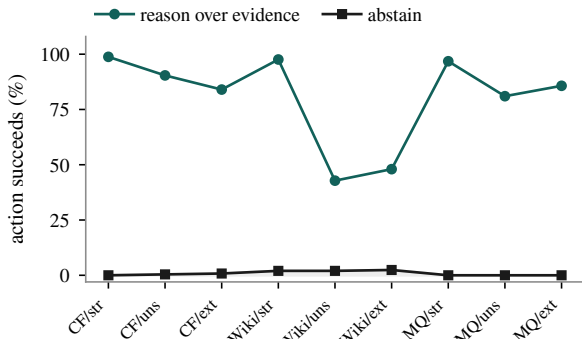


Figure 5: Per-action success rates. Abstention lies against the floor in every cell, which is the mechanism behind the zero headroom in Figure 4.

We therefore trained a head predicting retrieval reliability from the *shape* of the candidate set — margins to the next best candidate, how many candidates clear threshold, neighbourhood entropy, subject diversity — on CounterFact and WikiUpdate with MQuAKE-CF held out entirely. It reached 0.956 AUROC on the held-out dataset. The mechanism appeared in the fitted weights rather than being asserted: 71.5% of coefficient mass sat on margin features, and the largest single weight was -1.43 on the best *competing* candidate’s score. Used adaptively it produced +15.9 points on two MQuAKE-CF cells ($p = 0.0019$, Holm-corrected [5]), with zero discordant pairs where fixed gating already worked.

6.2 Measuring the ceiling

Rather than continue improving the router, we measured its ceiling. For 1,689 queries across all three datasets and conditions we executed *every* candidate action and scored the result, yielding per-query ground truth for which actions work.

Dataset	Condition	n	Static	Oracle
CounterFact	structured	250	1.0000	1.0000
CounterFact	unstructured	250	0.9040	0.9040
CounterFact	extracted	250	0.8400	0.8400
WikiUpdate	structured	250	0.9960	0.9960
WikiUpdate	unstructured	250	0.4280	0.4280
WikiUpdate	extracted	250	0.4800	0.4800
MQuAKE-CF	structured	63	1.0000	1.0000
MQuAKE-CF	unstructured	63	0.8095	0.8095
MQuAKE-CF	extracted	63	0.8571	0.8571
Pooled		1689	0.7874	0.7874

Table 1: Headroom is +0.0000 in every cell. “Static” is one line of code with no model and no training; “Oracle” is an upper bound no real router can exceed.

An oracle router choosing the best available action on every query is exactly equal to a one-line static policy — recite directly where legal, otherwise reason over the retrieved evidence — to four decimal places in all nine cells (Figure 4, Table 1). The maximum attainable gain of any per-query routing method over one line of code is 0.00 points.

The mechanism is stark (Figure 5). Abstention succeeded on 19 of 1,689 queries, and on all 19 of those, reasoning or direct recitation succeeded too. **Abstention is the sole winning action zero times.**

6.3 What this does to the preceding subsection

The gains in §5.1 are real and we stand behind the numbers. The *explanation* was wrong. We framed the head as adapting per query to the local regime; it does not, because there is no per-query decision worth making. What it does is suppress gates that misfire — it converges toward the static policy. Its gains measure the harm the fixed gates were doing, not insight it adds. A one-line rule matches it everywhere, with no training data and none of the k -fold cross-encoder cost.

7 Why Abstention Cannot Be Measured Here

The reason is structural. These are **counterfactual** benchmarks: the evaluation question asks for the *post-edit* answer, so answering from parametric knowledge returns the *pre-edit* value, which is wrong by construction. No query in the suite has “no edit applies, answer from what you know” as its correct behaviour.

This generalises beyond our system. Scope classification is load-bearing for the SERAC-descendant family [12], whose premise is a learned decision about whether a stored

edit applies. On these benchmarks that decision is degenerate: the answer is always “yes.” **A benchmark without negatives cannot measure a classifier’s ability to reject**, and an abstention path evaluated on one can only lose points.

It also explains a result of ours that had gone unexplained. On RippleEdits, which *does* contain same-subject different-relation queries and therefore partial negatives, INLAY’s preservation score was the worst in the comparison at 0.05, from keys over-firing. Scope precision matters exactly where a benchmark has negatives, which is precisely where INLAY is weakest and where the main benchmarks are silent.

The missing condition is constructible: ask evaluation questions against an index that does *not* contain the corresponding edit. On such queries abstention becomes correct, the oracle should separate from the static policy, and routing acquires measurable headroom for the first time.

8 Limitations

Recitation, not reasoning. Logit-space playback reproduces a stored answer; it cannot combine that answer with anything else, because by the final decoding step no reasoning remains to influence. This is structural.

Scope precision on hard negatives is the real open problem, as the RippleEdits score above indicates.

Small cells. MQuAKE-CF slices are $n = 63$. A +6.4-point structured-mode result rests on four discordant pairs, where the smallest attainable exact McNemar p is 0.125 — suggestive, not established.

One retrieval stack. A single encoder and one cross-encoder verifier throughout; sensitivity to those choices is untested.

A scale anomaly we diagnose but do not fix. WikiUp-date is 4.38 points *worse* at 7B than at 1.5B. It is not an editing failure: the larger model refuses to answer far more often on noisy evidence (46.88% vs 29.38%; 91.11% vs 66.67% where retrieval is wrong), and substring accuracy scores a refusal identically to a wrong answer. The larger model is more accurate whenever it commits — it simply commits less. This is a property of the metric, and it implies scale comparisons on noisy-retrieval benchmarks systematically understate larger models.

9 Conclusion

Freezing the weights and placing edits in an external addressable memory buys writes roughly $1600\times$ cheaper, exact deletion, and locality by construction. It pays for them by reciting facts rather than reasoning with them.

The result we did not expect to report is that a one-line static policy is provably optimal on all 1,689 queries we tested, because abstention can never be correct when every

question targets an edited fact. Reported routing gains on these benchmarks measure how much harm a gate was doing. An entire architectural family is being evaluated on suites that cannot reward its central component, and the remedy is not a better router but a benchmark containing queries for which the correct answer is “this edit does not apply.”

Reproducibility. All numbers come from logged runs. Splits are subject-disjoint with seed 0; the held-out dataset is never touched during training of any learned component. Code, per-experiment write-ups including negative results, and the figure generator are released with this paper.

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